INCH-POUND
MIL-PRF-28750/10E
28 September 2020
SUPERSEDING
MIL-PRF-28750/10D
7 July 2015

PERFORMANCE SPECIFICATION SHEET

RELAY, SOLID-STATE, SEALED, CLASS I, OPTICALLY ISOLATED, ZERO VOLTAGE TURN-ON, 25 AMPERES, 250 V MAXIMUM, 400 HZ, POWER SWITCHING, SPST (N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and MIL-PRF-28750.

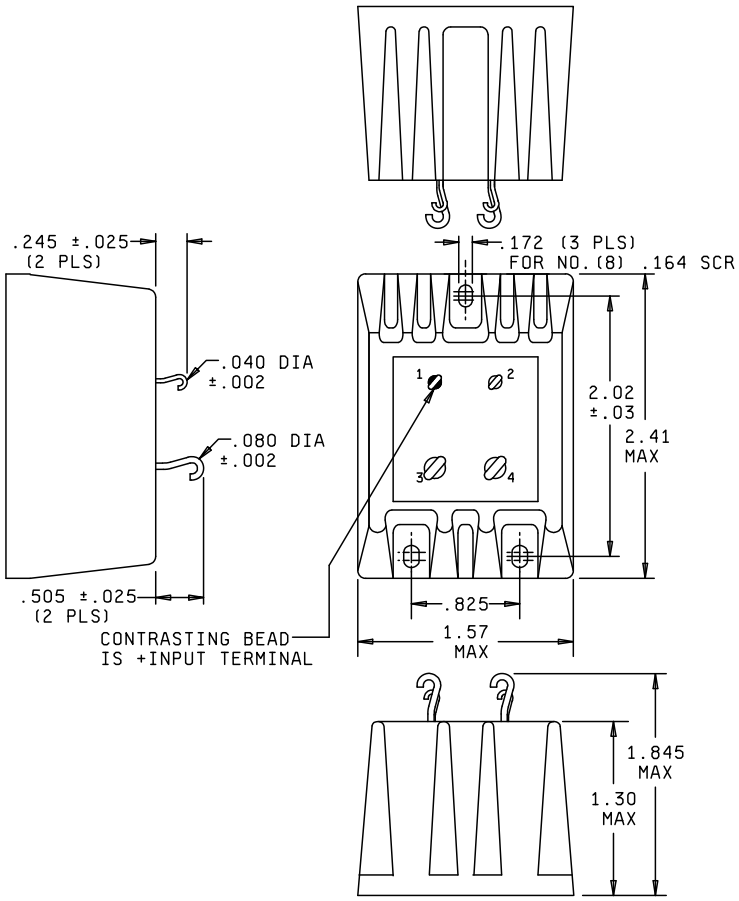
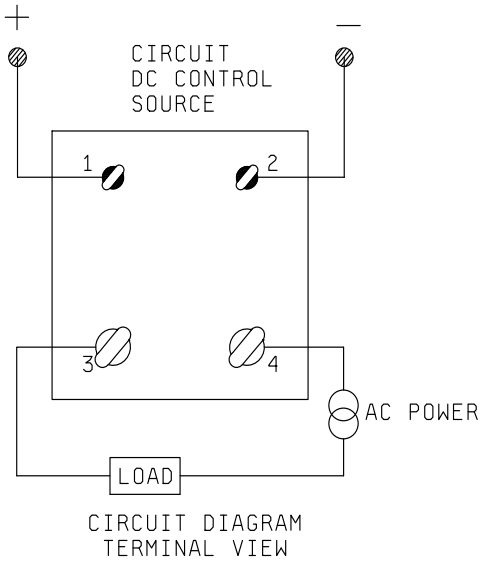


FIGURE 1. Outline drawing and dimensions.

AMSC N/A

FSC 5945

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Inches	Millimeters
.002	.051
.025	.635
.030	.762
.040	1.016
.080	2.032
.172	4.369
.245	6.223
.505	12.827
.550	13.970
.825	20.955
1.300	33.020
1.570	39.878
1.845	46.863
1.990	50.546
2.050	52.070
2.410	61.214

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (.25mm).
4. Terminal numbers shown above are for reference only, and do not appear on the header.

FIGURE 1. Outline drawing and dimensions - Continued.

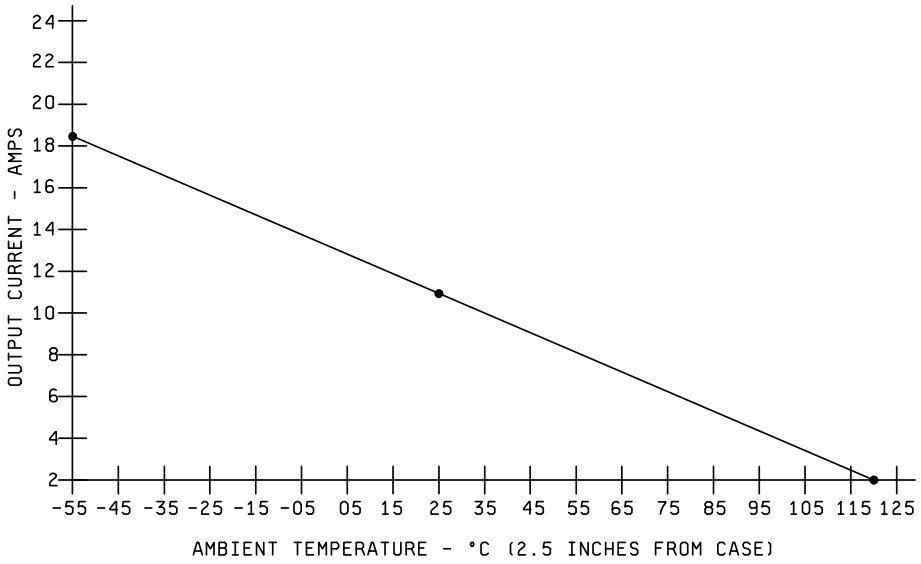


FIGURE 2. Output current verses ambient temperature.

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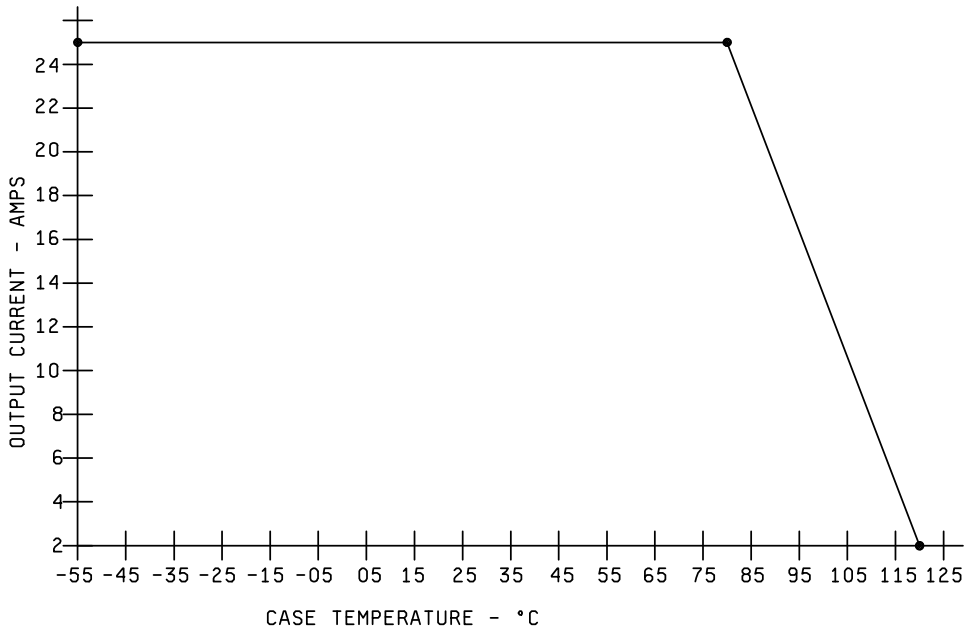


FIGURE 3. Output current versus case temperature.

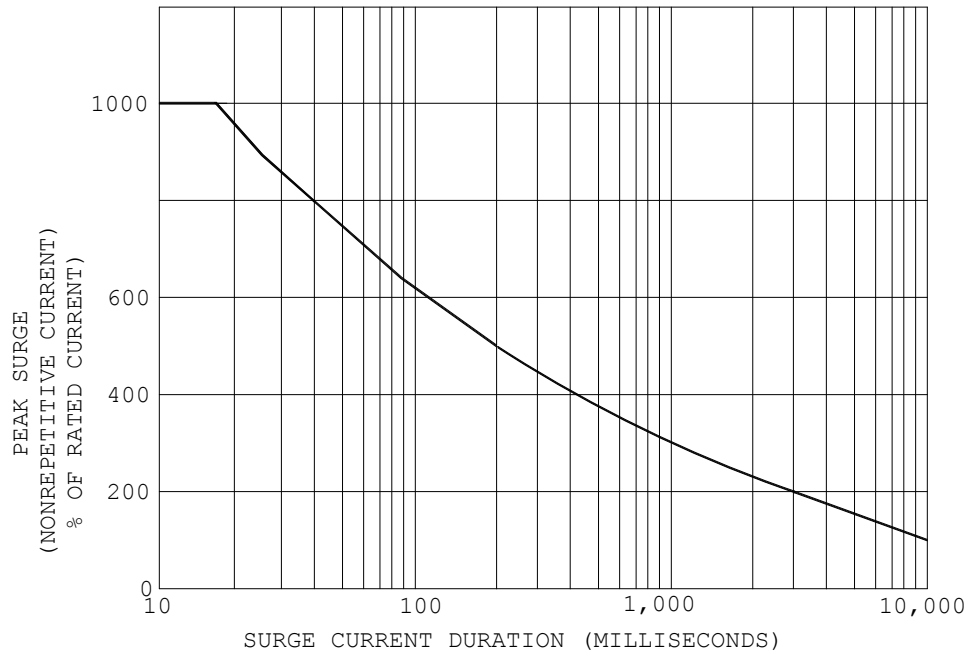


FIGURE 4. Peak surge current vs. surge current duration.

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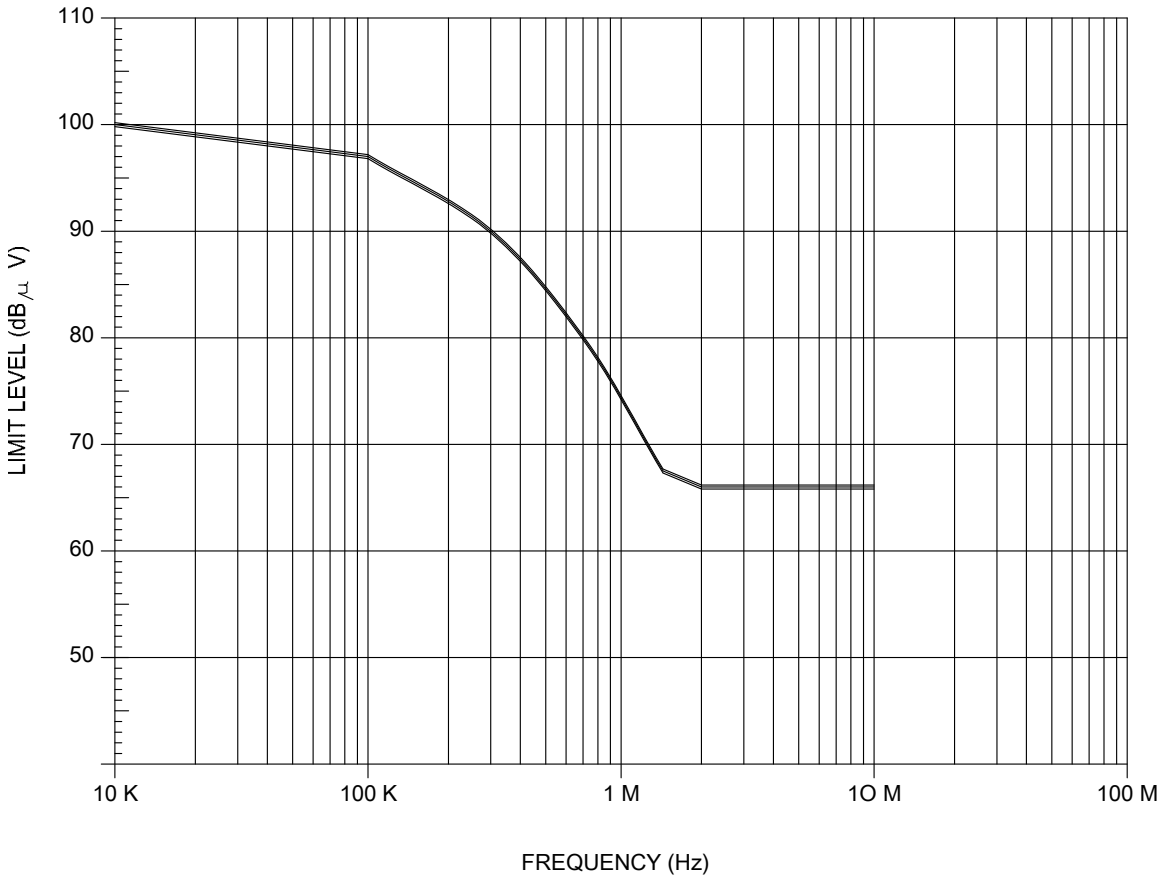


FIGURE 5. Conducted emissions – steady state condition.

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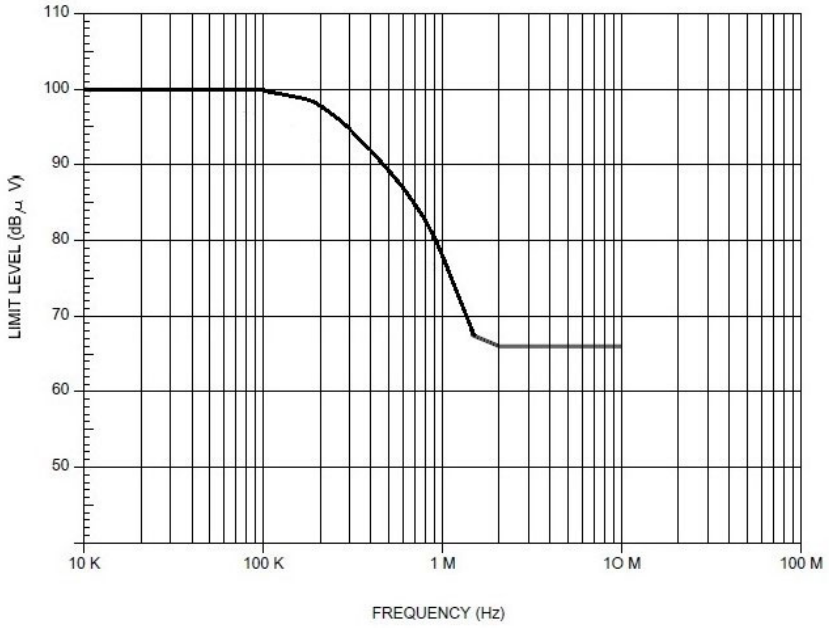


FIGURE 6. Conducted emissions – switching condition.

REQUIREMENTS:

SCREENING: Mechanical shock or constant acceleration shall be in accordance with MIL-PRF-28750, except the constant acceleration shall be performed at 100 g's.

MECHANICAL REQUIREMENTS:

Weight: 6 ounces maximum.

Seal: (Solder or weld) MIL-STD-202-112, test condition A, except temperature +125°C +0°C, -5°C.

Dimensions and configuration: See figure 1.

Terminals:

Terminal strength: 5 pounds pull.

Terminal solderability: Applicable.

Terminal finish: Corrosion resistant material. Corrosion resistant material such as gold plating are considered acceptable.

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INPUT REQUIREMENTS:

Input voltage range: 4.0 V dc to 32.0 V dc.

Assured turn-on voltage: ≤ 4.0 V dc.

Assured turn-off voltage: ≥ 1.0 V dc.

Input current: 16 mA maximum at 32 V dc.

Turn-on time: $\frac{1}{2\,(\text{line frequency})}$ maximum.

Turn-off time: $\frac{1}{(\text{line frequency})}$ maximum.

Bias current: Not applicable.

OUTPUT REQUIREMENTS:

Output voltage range: 25 V ac to 250 V ac, 45 Hz to 440 Hz.

Rated output current: 25 A, ac maximum (see figure 2 and figure 3).

Rated output voltage: 220 V maximum,400 Hz.

Output voltage drop: 1.5 V rms maximum.

Output leakage current: 10 mA, ac maximum at 220 V ac, 400 Hz.

Crosstalk: Not applicable.

Transient voltage: Step line voltage from 230 Volts AC, 400 Hz to 360 Volts AC, 400 Hz. For 5 seconds minimum.

Electric system spike: Not applicable.

Overload: 80 amperes.

DC offset voltage: ±150 mV maximum.

Waveform distortion: 4 V rms maximum from 10 percent to 100 percent rated output current.

Minimum current: Not applicable.

Zero crossover voltage (at +25°C):

M28750/10-001 = 0.0 ± 15 volts peak maximum.
M28750/10-002 = 0.0 ±40 volts peak maximum.

ELECTRICAL REQUIREMENTS:

Dielectric withstanding voltage: 1,500 V ac rms at 60 Hz.

Insulation resistance: 100 megohms minimum.

Isolation: 10 picofarads maximum.

Power dissipation: 38 watts maximum.

Exponential rate of voltage rise (dv/dt): 100 volts/μsecond.

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Electromagnetic interference: Maximum broadband conducted emission on power lines with 115 V ac, 400 Hz, 10 amperes resistive load during steady state and switching conditions (see Figures 5 and 6).

ENVIRONMENTAL REQUIREMENTS:

Temperature:

Operation: -55°C to +110°C.

Storage: -55°C to +125°C.

Shock (specified pulse): MIL-STD-202-213, test condition F (1500 g's).

Vibration: 30 g's, 10 Hz to 3,000 Hz.

Moisture resistance: Not applicable.

Resistance to soldering heat: MIL-STD-202-210, test condition A.

Salt atmosphere (corrosion): Shall be in accordance with MIL-PRF-28750, except examination after test shall be in accordance with method 1041 of MIL-STD-750.

Qualification: To qualify either M28750/10-001 or M28750/10-002 all qualification inspections of MIL-PRF-28750 shall be performed. In order to qualify M28750/10-001 and M28750/10-002 submit M28750/10-001 to qualification inspection of MIL-PRF-28750 as applicable and submit M28750/10-002 (2 units) to group A inspection of MIL-PRF-28750.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-28750, this document references the following:

MIL-STD-202-213 MIL-STD-202-210 MIL-STD-202-112 MIL-STD-750

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-058)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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